Waveform-based timing correction for time-of-flight positron emission tomography

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Abstract

Accurate coincidence timing is central to time-of-flight positron emission tomography. This study describes a waveform-based correction of leading-edge timing for two scintillation-detector modules using the same crystals and photodetectors with alternative analog front-end electronics. Waveforms are acquired at two different stages of cascaded amplification and used to estimate event-specific timing corrections. The analysis examines amplification stage, detector sharing, estimator family and waveform-window extent.

1 Introduction

Positron emission tomography (PET) detects pairs of annihilation photons to reconstruct radiotracer distributions. Time-of-flight (TOF) information uses the difference between photon arrival times to constrain the annihilation position along the line joining the detectors and can improve image signal-to-noise performance. The precision of this information depends on the coincidence timing resolution (CTR), conventionally expressed as the full width at half maximum (FWHM) of the time-difference distribution [1].

A leading-edge discriminator (LED) assigns an arrival time when a pulse crosses a fixed amplitude threshold. Variations in pulse amplitude and rising-edge shape shift the crossing time, producing time walk. Digitized waveforms retain information about these variations, which machine learning (ML) can exploit to estimate event-specific timing corrections.

Waveform-based approaches include direct TOF estimation and correction of conventional timing measurements. Berg and Cherry demonstrated direct TOF estimation from detector waveform pairs using a convolutional neural network (CNN), improving CTR by 20% relative to LED, from 231 to 185 ps [2]. Onishi et al. combined LED with a CNN trained using waveforms acquired at a single source position, improving timing resolution while avoiding the position-dependent bias associated with direct TOF estimation [3]. Feng et al. explored convolutional features combined with transformer-based attention to capture local and global temporal information [4]. For bismuth germanate (BGO) detectors, Loignon-Houle et al. compared LED, a two-threshold time-walk correction and CNN-based timing, showing that performance gains and timing-distribution tails depend on detector characteristics [5].

An open question is whether waveform samples beyond the rising edge improve ML-based LED timing corrections, and which temporal regions contribute most. Previous CNN studies have primarily examined short windows near the rising edge [3, 5], leaving extended-window performance at high sampling rates insufficiently characterized. In related model-based work, Ruiz-Gonzalez et al. observed diminishing returns from additional samples and found that the advantage of maximum-likelihood timing over digital discrimination disappeared above 2 GS/s [6]. These results do not establish whether extending the waveform window benefits data-driven corrections at the 80 GS/s acquisition rate used here.

The analog readout chain is another determinant of timing performance. Amplification can change signal amplitude, bandwidth, noise and rise time, affecting both LED timing and the waveform features accessible to ML. Pourashraf et al. compared front-end readout configurations for silicon photomultiplier (SiPM)-based TOF-PET detectors, reporting a best CTR of 99.4 ± 1.9 ps, more than 20 ps better than two alternative configurations [7]. These results establish the importance of electronics choices, but do not determine whether ML corrections applied to outputs of successive stages of cascaded amplification perform differently.

Building on previous waveform-based timing studies, this work investigates how different ML approaches exploit detector signals to correct leading-edge timing, rather than focusing exclusively on achieving the lowest CTR. By comparing multiple estimator families and temporal windows, we aim to characterize the timing information available across the waveform and assess the potential of both conventional and dedicated high-gain readouts.

A further objective is to investigate the scalability of ML-based timing calibration. Training a separate model for every detector pair would become impractical in complex PET systems. We therefore introduce detector-shared estimators, which apply the same correction function to both detector waveforms, enforcing antisymmetry under detector exchange. Comparing these models with unconstrained estimators allows us to assess whether detector-specific relationships are necessary or whether general waveform characteristics are sufficient for effective timing correction.

Although complete waveform acquisition may not be practical in clinical systems, this analysis provides a way to investigate which signal characteristics ML models exploit and which temporal regions are most relevant to their predictions. The two electronics boards are evaluated independently using control, development, and blind acquisitions, with temporal sensitivity analysis complementing the timing-performance evaluation.

2 Experimental setup and datasets

2.1 Detector modules and acquisition

Two opposing detector modules consist of silicon photomultipliers (SiPMs) coupled to $2 \times 2 \times 3\,\mathrm{mm}^3$ lutetium–yttrium oxyorthosilicate (LYSO) crystals, with a ^{22}Na source posi-

tioned centrally between them. The University of Chicago (UC) and Fondazione Bruno Kessler (FBK) configurations use the same crystals and SiPMs but different analog front-end electronics.

The two-stage readout chain is shown in Figure 1. Its first stage produces a relatively long signal suitable for energy estimation and conventional timing. The second provides additional amplification and a sharper signal optimized for timing. This dedicated stage requires a separate 5 V supply, adding hardware and integration challenges that make it chalenging to deploy in clinical PET systems. The two outputs are referred to as *energy* and *timing*, respectively.

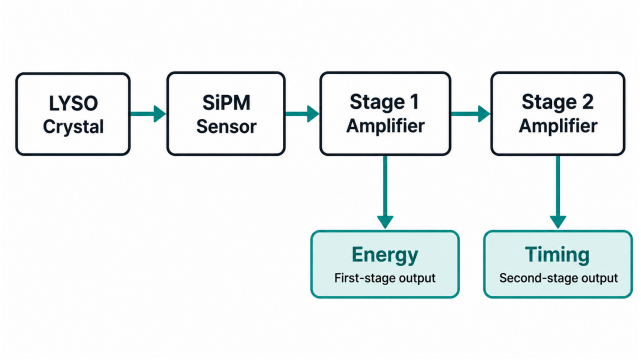


Figure 1: Two-stage readout chain. Energy and timing waveforms are sampled from successive amplification stages of the same SiPM pulse.

The energy signal is acquired without amplitude clipping, and its maximum is used for photopeak selection. For the high-gain timing signal, the oscilloscope's narrower vertical range emphasizes the rising edge while clipping the pulse peak, following a strategy also used by Onishi et al. [3]. In these acquisitions, the energy waveform is cleaner and more stable but exhibits greater time walk; the timing waveform is more susceptible to noise and pulse-shape variations.

2.2 Acquisitions and scientific roles

Each coincidence event contains four simultaneous waveforms: first-stage outputs from detectors 1 and 2 on channels 1 and 2, and second-stage outputs on channels 3 and 4. The 100 ns records are sampled at 80 GS s^{-1} with a SiPM bias of 48 V. The centrally located source defines a true TOF of zero.

For each board, three independent acquisitions are used: *calibration* to determine preprocessing and LED selection criteria, *train* for model training and selection, and *test* for final evaluation. Each calibration dataset contains approximately 20,000 raw coincidence events, while the train and test sets contain approximately 100,000 events each, before event selection. The two boards are analyzed independently.

Figures 2 and 3 show the energy and timing waveforms of the same illustrative coincidence event for UC and FBK, respectively. Both detector signals are shown for each amplification stage. These are raw waveforms before event selection or preprocessing, with only the acquisition vertical-range clipping applied.

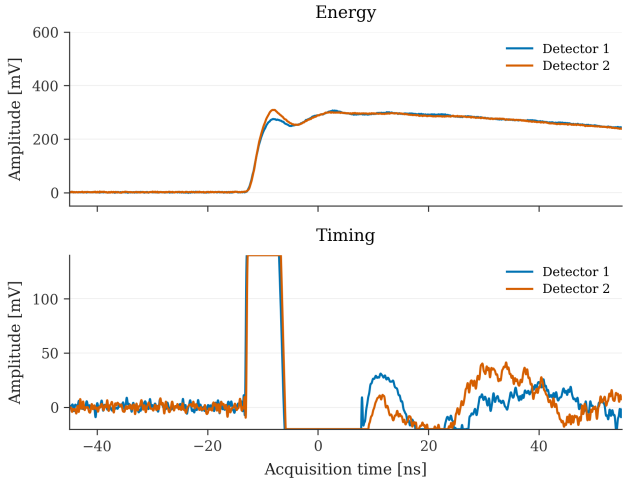


Figure 2: UC raw waveforms: first-stage (energy) and second-stage (timing) signals from both detectors in one event.

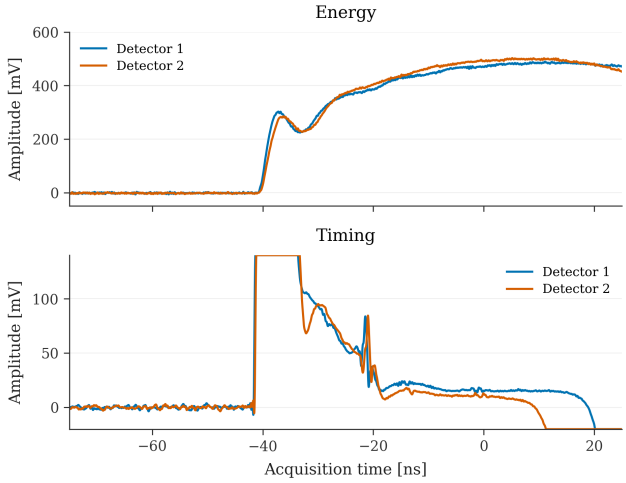


Figure 3: FBK raw waveforms: first-stage (energy) and second-stage (timing) signals from both detectors in one event.

3 Methods

3.1 Event selection and LED calibration

Only photopeak events are retained, based on the maximum amplitudes of both first-stage (energy) waveforms. Acceptance intervals are obtained from iterative Gaussian fits to the photopeaks in the calibration acquisition and applied to both waveform modes. For timing-mode analysis, time-over-threshold (ToT) selection additionally rejects short, spurious crossings associated with noise and signal oscillations. Events with excessive pre-pulse baseline fluctuations or baseline samples near the acquisition limits are excluded. All selection criteria are established from calibration data only.

For each mode, a leading-edge discriminator (LED) threshold is chosen on the calibration acquisition to minimize coincidence timing resolution (CTR), subject to a minimum fraction of valid coincidences within the prescribed time range. A fixed channel-pair timing offset is subsequently estimated from the calibration LED residuals. Both the threshold and offset remain unchanged in training and testing.

3.2 Timing-error model and learning target

The goal is to correct waveform-dependent errors in a conventional LED measurement, rather than estimate an unconstrained time of flight directly. For a coincidence event, let $\Delta t_{\mathrm{LED}} = t_{1,\mathrm{LED}} - t_{2,\mathrm{LED}}$. We model this measurement as

$$\Delta t_{\mathrm{LED}} = \tau + C + \delta(s_1, s_2) + \epsilon, \tag{1}$$

where τ is the true time of flight, C is the fixed channel-pair offset, δ denotes the waveform-dependent timing error, and ϵ represents residual unpredictable variation. The waveform-dependent component δ is the quantity we aim to estimate and correct using ML.

Writing $\widehat{C}$ for the fixed offset estimated during calibration, the supervised target and corrected time difference are

$$y = \Delta t_{\mathrm{LED}} - \tau - \widehat{C}, \tag{2}$$

$$\widehat{\Delta t} = \Delta t_{\mathrm{LED}} - \widehat{C} - h_\theta(s_1, s_2). \tag{3}$$

The final residual relative to true TOF is:

$$r = \widehat{\Delta t} - \tau = y - h_\theta(s_1, s_2). \tag{4}$$

This formulation separates waveform-dependent timing errors, which can potentially be corrected, from unpredictable noise. The source is positioned centrally between the detectors, giving $\tau = 0$. Any positioning error would introduce a constant offset, which can be absorbed into the calibration term without affecting CTR.

Energy and timing waveforms are analyzed separately. Each input waveform is expressed relative to its own LED crossing and normalized using fixed channel-specific amplitude limits. This alignment removes absolute acquisition time from the input. Events lacking a valid coincidence or the full requested window are excluded.

3.3 Detector-shared and direct corrections

Previous waveform-based timing studies have primarily used direct estimators, which predict the timing correction from the pair of detector waveforms [2, 3]. These models can learn detector-specific relationships without imposing constraints on how the correction changes when the detector inputs are exchanged.

While effective in a two-detector setup, training a separate model for every detector pair would be difficult to scale to a complete PET system. This motivates a detector-shared formulation, in which the correction is expressed as the difference between two evaluations of the same function:

$$h_\theta(s_1, s_2) = g_\theta(s_1) - g_\theta(s_2). \tag{5}$$

The shared function g_θ estimates a waveform-dependent timing contribution for each detector independently. This formulation enforces detector-exchange antisymmetry: swapping the inputs reverses the correction sign, while identical inputs yield zero correction. It also offers a potentially more scalable approach, as the same model can be applied to different detector pairs without requiring pair-specific parameters.

Comparing direct and shared estimators allows us to investigate whether effective timing corrections require detector-specific relationships or can be captured by a common waveform function. Different model families, including linear regression, multilayer perceptrons (MLPs), convolutional neural networks (CNNs), locally connected networks and MiniRocket-based estimators, are considered to assess the influence of model architecture. Table 1 summarizes the estimators, while implementation details are provided in the appendix.

3.4 Model comparison and training protocol

Model selection takes place exclusively within the training acquisition, using three common cross-validation folds for estimators subject to outer cross-validation. The overall protocol is summarized in Figure 4. The calibration acquisition fixes the preprocessing and LED-calibration rules, the training acquisition is used for model selection and final fitting, and the test acquisition is reserved for independent evaluation.

Linear ridge regression is the methodological exception: regularization is chosen by internal leave-one-out cross-validation minimizing mean squared error, and the resulting model is fitted on the full training population without the outer CTR-based selection. MiniRocket-based ridge estimators follow the ordinary outer cross-validation protocol. Hyperparameter search spaces and training specifications are documented in Appendix A.

The calibration, training, and test acquisitions are independent: the calibration acquisition fixes signal-selection and reference parameters, the training acquisition determines supervised model parameters, and the test acquisition is reserved for final performance characterization.

Formulation	Estimator	Correction structure
Direct	Linear ridge	Joint linear regression on concatenated waveforms.
	Dense MLP	Joint nonlinear regression on concatenated waveforms.
	Independent CNN	Separate detector-specific CNN scorers; output difference.
	Onishi-inspired CNN	Paired waveforms fused by convolution for joint regression.
	MiniRocket + ridge	Multivariate features of the waveform pair; ridge regression.
Shared	Linear ridge	Linear regression on the difference of waveform samples.
	Dense MLP	Common nonlinear detector scorer; output difference.
	Locally connected MLP	Common position-specific local scorer; bounded output difference.
	Temporal CNN	Common convolutional detector scorer; output difference.
	MiniRocket + ridge	Common waveform feature transform; feature difference and ridge regression.

Table 1: Waveform-based timing estimators implemented in the benchmark. Direct estimators do not enforce detector-exchange antisymmetry; shared estimators use a common detector-level function or feature transform to enforce it.

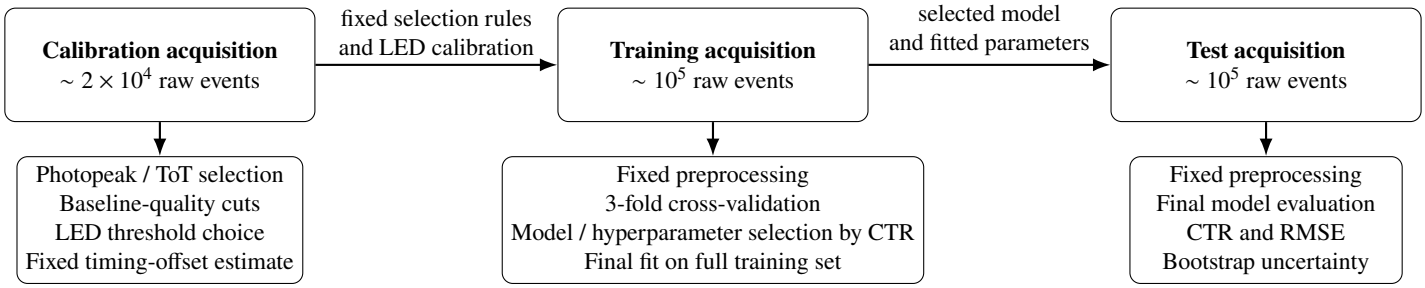


Figure 4: Schematic overview of the experimental protocol. The calibration acquisition is used to determine fixed event-selection and LED-calibration criteria. Model selection and fitting are performed on the independent training acquisition. Final performance is assessed on the independent test acquisition.

3.5 Timing performance and independent evaluation

All configurations are evaluated against LED on the same selected test events. CTR is defined as the full width at half maximum (FWHM) of the timing-residual distribution and computed using histogram-based method F1 of Rainio et al. [8]. Identical histogram settings are used for LED and corrected residuals within each mode. The fixed bin widths are 10 ps for energy and 5 ps for timing.

Root mean squared error (RMSE) relative to true TOF complements CTR because it captures both residual dispersion and systematic bias. In particular, $\mathrm{RMSE}^2 = \mathrm{Var}(r) + [\mathbb{E}(r)]^2$; a reduction in RMSE need not imply a comparable narrowing of the central timing peak. Both metrics are therefore needed to characterize the effects of calibration and waveform-based correction.

Fixed preprocessing and the fitted model are applied to the independent test acquisition. Test-event bootstrap resampling provides uncertainties on CTR and RMSE without retraining; these estimates are conditional on the selected fitted model. Within-board comparisons use matched test events and joint resampling to assess paired model differences. The two electronics boards are characterized separately rather than treated as a direct between-board performance ranking.

3.6 Temporal sensitivity analysis

To investigate which parts of each waveform influence the correction, temporal occlusion is applied to the trained models on test events. Short groups of samples are replaced by interpolation from neighboring samples, and their importance is quantified by the average absolute change in predicted correction. Shared estimators are perturbed synchronously in both detector inputs, while direct estimators are perturbed one input at a time.

The resulting profiles reveal whether predictions depend primarily on the rising edge or also on later waveform characteristics, and allow comparisons across windows and amplification stages. They measure sensitivity to the chosen perturbation rather than causal or uniquely attributable timing information, particularly because adjacent samples are strongly correlated.

4 Results

4.1 University of Chicago board

[Placeholder: UC validation and blind CTR by estimator, waveform mode and window (Tables 2 and 3; Figure 5).]
[Placeholder: UC temporal sensitivity by estimator and waveform window (Figure 6).]

Placeholder: UC within-board CTR comparison by model and window.

Figure 5: UC within-board CTR comparison by model and window.

Formulation	Estimator	Validation CTR [ps]	Blind CTR [ps]	Blind LED CTR [ps]
Shared	MiniRocket	—	—	—
	Linear ridge	n/a	—	—
	Dense MLP	—	—	—
	Locally connected MLP	—	—	—
	Temporal CNN	—	—	—
Direct	MiniRocket	—	—	—
	Linear ridge	n/a	—	—
	Dense MLP	—	—	—
	Onishi-inspired paired CNN	—	—	—
	Independent temporal CNN	—	—	—

Table 2: UC, timing waveforms, -1 to 2 ns input interval. Validation CTR is summarized by the outer-CV mean and fold dispersion; blind CTR uses F1 with bootstrap uncertainty. Outer validation is not applicable to linear ridge (n/a).

Formulation	Estimator	Validation CTR [ps]	Blind CTR [ps]	Blind LED CTR [ps]
Shared	MiniRocket	—	—	—
	Linear ridge	n/a	—	—
	Dense MLP	—	—	—
	Locally connected MLP	—	—	—
	Temporal CNN	—	—	—
Direct	MiniRocket	—	—	—
	Linear ridge	n/a	—	—
	Dense MLP	—	—	—
	Onishi-inspired paired CNN	—	—	—
	Independent temporal CNN	—	—	—

Table 3: UC, timing waveforms, -2 to 30 ns input interval. Validation CTR is summarized by the outer-CV mean and fold dispersion; blind CTR uses F1 with bootstrap uncertainty. Outer validation is not applicable to linear ridge (n/a).

Placeholder: UC temporal-occlusion sensitivity maps.

Figure 6: UC temporal-occlusion sensitivity maps.

Placeholder: FBK temporal-occlusion sensitivity maps.

Figure 8: FBK temporal-occlusion sensitivity maps.

4.2 Fondazione Bruno Kessler board

[Placeholder: FBK validation and blind CTR by estimator and window: energy waveforms (Tables 4 and 5) and timing waveforms (Tables 6 and 7), with the within-board comparison in Figure 7.]
[Placeholder: FBK temporal sensitivity by estimator and waveform window (Figure 8).]

Placeholder: FBK within-board CTR comparison by model and window.

Figure 7: FBK within-board CTR comparison by model and window.

Formulation	Estimator	Validation CTR [ps]	Blind CTR [ps]	Blind LED CTR [ps]
Shared	MiniRocket	—	—	—
	Linear ridge	n/a	—	—
	Dense MLP	—	—	—
	Locally connected MLP	—	—	—
	Temporal CNN	—	—	—
Direct	MiniRocket	—	—	—
	Linear ridge	n/a	—	—
	Dense MLP	—	—	—
	Onishi-inspired paired CNN	—	—	—
	Independent temporal CNN	—	—	—

Table 4: FBK, energy waveforms, -1.5 to 2 ns input interval. Validation CTR is summarized by the outer-CV mean and fold dispersion; blind CTR uses F1 with bootstrap uncertainty. Outer validation is not applicable to linear ridge (n/a).

Formulation	Estimator	Validation CTR [ps]	Blind CTR [ps]	Blind LED CTR [ps]
Shared	MiniRocket	—	—	—
	Linear ridge	n/a	—	—
	Dense MLP	—	—	—
	Locally connected MLP	—	—	—
	Temporal CNN	—	—	—
Direct	MiniRocket	—	—	—
	Linear ridge	n/a	—	—
	Dense MLP	—	—	—
	Onishi-inspired paired CNN	—	—	—
	Independent temporal CNN	—	—	—

Table 5: FBK, energy waveforms, -2 to 30 ns input interval. Validation CTR is summarized by the outer-CV mean and fold dispersion; blind CTR uses F1 with bootstrap uncertainty. Outer validation is not applicable to linear ridge (n/a).

Formulation	Estimator	Validation CTR [ps]	Blind CTR [ps]	Blind LED CTR [ps]
Shared	MiniRocket	—	—	—
	Linear ridge	n/a	—	—
	Dense MLP	—	—	—
	Locally connected MLP	—	—	—
	Temporal CNN	—	—	—
Direct	MiniRocket	—	—	—
	Linear ridge	n/a	—	—
	Dense MLP	—	—	—
	Onishi-inspired paired CNN	—	—	—
	Independent temporal CNN	—	—	—

Table 6: FBK, timing waveforms, -1.5 to 2 ns input interval. Validation CTR is summarized by the outer-CV mean and fold dispersion; blind CTR uses F1 with bootstrap uncertainty. Outer validation is not applicable to linear ridge (n/a).

Formulation	Estimator	Validation CTR [ps]	Blind CTR [ps]	Blind LED CTR [ps]
Shared	MiniRocket	—	—	—
	Linear ridge	n/a	—	—
	Dense MLP	—	—	—
	Locally connected MLP	—	—	—
	Temporal CNN	—	—	—
Direct	MiniRocket	—	—	—
	Linear ridge	n/a	—	—
	Dense MLP	—	—	—
	Onishi-inspired paired CNN	—	—	—
	Independent temporal CNN	—	—	—

Table 7: FBK, timing waveforms, -2 to 30 ns input interval. Validation CTR is summarized by the outer-CV mean and fold dispersion; blind CTR uses F1 with bootstrap uncertainty. Outer validation is not applicable to linear ridge (n/a).

5 Conclusions

[Placeholder: Within-board findings and qualitative differences in model, waveform-mode, window and temporal-sensitivity behaviour between the electronics boards.]

A Hyperparameter spaces and model libraries

The configured model spaces and training settings are listed for dense and locally connected estimators in Table 8, temporal CNNs in Table 9, the paired CNN in Table 10, and linear and MiniRocket estimators in Table 11. Square brackets denote hidden-layer widths or convolutional channel sequences; braces denote alternative grid values. A stated range with a count denotes logarithmic spacing. All model comparisons use the same input intervals within each board.

The model libraries and supporting scientific software are listed in Table 12. The listed version bounds are declared minimum requirements.

Formulation	Estimator	Parameter	Configured space or value
Shared	Dense MLP	Dense hidden widths	{[16], [128], [128, 64], [16, 16, 16]}
		Activation	ReLU
		Learning rate	$\{0.01, 0.001, 10^{-4}, 10^{-5}\}$
		Batch size	32
		Optimizer	Nesterov SGD; momentum 0.9
		Maximum epochs	300
		Internal holdout fraction	0.2
		Loss-stop patience [epochs]	10
		Minimum loss improvement [ps]	0.1
		Gradient clipping norm	10
		Gradient-stop threshold	10^{-6}
		Gradient-stop patience [epochs]	4
	Locally connected MLP	Dense hidden widths	[128]
		Activation	ReLU
		First local kernel size	3
		First local stride	1
		Second local kernel size	3
		Second local stride	1
		Smooth correction bound [ps]	500
		Learning rate	$\{0.01, 0.001, 10^{-4}, 10^{-5}\}$
		Batch size	{32, 128}
		Optimizer	Adam
		Maximum epochs	300
		Internal holdout fraction	0.2
		Loss-stop patience [epochs]	10
		Minimum loss improvement [ps]	0.1
		Gradient clipping norm	10
		Gradient-stop threshold	10^{-6}
		Gradient-stop patience [epochs]	4
Direct	Dense MLP	Dense hidden widths	{[16], [128], [128, 64], [16, 16, 16]}
		Activation	ReLU
		Learning rate	$\{0.01, 0.001, 10^{-4}, 10^{-5}\}$
		Batch size	32
		Optimizer	Adam
		Maximum epochs	300
		Internal holdout fraction	0.2
		Loss-stop patience [epochs]	10
		Minimum loss improvement [ps]	0.1
		Gradient clipping norm	10
		Gradient-stop threshold	10^{-6}
		Gradient-stop patience [epochs]	4

Table 8: Dense and locally connected neural model spaces. Neural losses use RMSE. Gradient-based stopping is enabled with the listed threshold and patience.

Formulation	Estimator	Parameter	Configured space or value
Shared	Temporal CNN	Dense hidden widths	[64]
		Activation	ReLU
		Learning rate	$\{0.001, 10^{-4}, 10^{-5}\}$
		Batch size	{32, 128}
		Convolutional channels	[16, 32]
		Temporal kernel sizes	[5, 3]
		Optimizer	Adam
		Maximum epochs	300
		Internal holdout fraction	0.2
		Loss-stop patience [epochs]	10
		Minimum loss improvement [ps]	0.1
		Gradient clipping norm	10
		Gradient-stop threshold	10^{-6}
		Gradient-stop patience [epochs]	4
Direct	Independent temporal CNN	Dense hidden widths	[64]
		Activation	ReLU
		Learning rate	$\{0.001, 10^{-4}, 10^{-5}\}$
		Batch size	{32, 128}
		Convolutional channels	[16, 32]
		Temporal kernel sizes	[5, 3]
		Optimizer	Adam
		Maximum epochs	300
		Internal holdout fraction	0.2
		Loss-stop patience [epochs]	10
		Minimum loss improvement [ps]	0.1
		Gradient clipping norm	10
		Gradient-stop threshold	10^{-6}
		Gradient-stop patience [epochs]	4

Table 9: Temporal CNN model spaces. Kernels are valid convolutions without pooling; the dense head follows flattened convolutional features. Neural losses use RMSE.

Formulation	Estimator	Parameter	Configured space or value
Direct	Onishi-inspired paired CNN	Convolutional channels	[32, 64]
		Temporal kernel sizes	[5, 3]
		Dense hidden width	256
		Learning rate	$\{0.001, 10^{-4}, 10^{-5}\}$
		Batch size	{32, 128}
		Optimizer	Adam
		Maximum epochs	100
		Learning-rate milestones [epochs]	[30, 60]
		Learning-rate decay factor	0.1

Table 10: Paired CNN settings. The first kernel spans both detectors; later kernels span one detector-height position. Training uses MSE on the complete fitting subset, without an internal early-stopping holdout.

Formulation	Estimator	Parameter	Configured space or value
Shared	Linear ridge	Regularization grid	50 logarithmically spaced values from 10^{-8} to 1000
		Internal selection	Leave-one-out CV; MSE
		Intercept	None
	MiniRocket	Requested MiniRocket features	10000
		Ridge regularization strengths	{0.1, 0.3, 1, 3, 10, 30, 100, 300, 1000, 3000, 10000, 30000, 100000, 300000, 1e+06}
		Feature fitting	Control acquisition; scaling without centring
		Regression intercept	None
		Selection	Outer CV; mean CTR
Direct	Linear ridge	Regularization grid	50 logarithmically spaced values from 10^{-8} to 1000
		Internal selection	Leave-one-out CV; MSE
		Intercept	Included
	MiniRocket	Requested MiniRocket features	10000
		Ridge regularization strengths	{0.1, 0.3, 1, 3, 10, 30, 100, 300, 1000, 3000, 10000, 30000, 100000, 300000, 1e+06}
		Feature fitting	Control acquisition; scaling without centring
		Regression intercept	Included
		Selection	Outer CV; mean CTR

Table 11: Linear and MiniRocket spaces. Only the two linear estimators use internal ridge CV. MiniRocket transforms are frozen before outer supervised validation.

Library	Declared minimum	Role
PyTorch	2.2	Dense, locally connected and convolutional neural networks; tensor differentiation and training.
scikit-learn	1.4	Ridge regression, internal ridge CV and feature scaling.
sktime	0.35	Univariate and multivariate MiniRocket transforms.
NumPy	1.26	Waveform arrays, timing operations and bootstrap sampling.
SciPy	1.11	Scientific numerical routines and Gaussian fits for event selection.
Matplotlib	3.8	Timing, comparison and temporal-sensitivity figures.
pandas	2.1	Tabular data and transformed features.
Optuna	4.0	Optional adaptive parameter proposal; supplied benchmark spaces use finite grids.

Table 12: Model libraries and supporting scientific software.

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